

Zhang Xichen

The Hong Kong Polytechnic University
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Education

Hong Kong Polytechnic University

M.S. Electric Vehicles

Qs world ranking : 54

2024.9 - 2026.4

- Graduate with credit
- GPA:3.08/4.30
- Dissertation: Advanced SVPFM for Three-level converter based wireless power transfer system

Nanjing Tech University

B.ENG Electronic Engineering (International cooperation)

Shanghai ranking of Chinese Universities : 89

2020.9 - 2024.6

- GPA:83/100 (3.3/4.0)
- Graduate thesis: Design of SVPWM motor control device based on ESP32

Professional Experience

2023 Dataarobtic Co. Ltd., Internship

2024 Sinognss Co. Ltd., Internship

Awards

2023 Raicom Robot Development Competition

The Ministry of Industry and Information Technology of China

Provincial Third Prize

2022 Undergraduate Academic and Technological Forum of Nanjing Tech University

Nanjing Tech University

Outstanding Award

China Collegiate Computing Contest

CERACU

Second Prize of East China

Qualification

2023 IELTS 6.5

Research Experience

SCA Motor Development Group Internship

Intern

Dataarobotics Co.Ltd

2023.7

- Engineered a custom motor communication PCB for robot arms by Alitum Designer utilizing the BiSS protocol.
- Programmed firmware in C for data parsing, and executed hardware debugging using oscilloscopes, heat tests were also conducted.

Hardware & Control Development in Raicom Robot Development Competition

Independent Developer

Nanjing Tech University

2023.4-2023.7

- Developed the hardware layout for the robotic motor and power systems, progressing from circuit design to actual physical integration including hardware selection, PCB design through Alitum Designer, and debugging.
- Programmed PID feedback loops to dynamically adjust microcontroller PWM outputs, ensuring accurate speed and torque regulation of DC motors.
- This project won the provincial third prize form the Ministry of Industry and Information Technology of China.

- Designed and manufactured a brush-less DC motor driver board that integrates a magnetic encoder for real-time position feedback and ESP32 controller.
- Developed the control system on ESP32 controller by C using SVPWM modulation to control DC motor.
- System functions including real-time force feedback, provides real-time motor position feedback via a magnetic encoder integrated into the motor driver PCB.

Patent Drafting: A PFM modulated Three-level inverter based wireless power transfer system

Hong Kong Polytechnic University

2025.3-2025.6

- Based on master's degree dissertation with improvement.
- Proposed a wireless power transmission system based on a PFM-based multilevel inverter, comprising a controller, a drive circuit, a sampling circuit, a DC voltage source, and a flying-capacitor-type n-level inverter circuit.
- Authored the patent application based on the developed A-SVPFM strategy and voltage balancing algorithm.

Advanced-SVPFM modulation for Three-level converter based wireless power transfer system Hong Kong Polytechnic University
2024.12-2025.11

Dissertation/Supervised by Prof. Liu Wei

- Derive, design and verified a new modulation method for 3-level DC-DC converter for wireless charging application as the advanced SVPFM strategy ; Also derived comprehensive mathematical models to achieve wide-range ZVS for half-bridge flying-capacitor three-level converters in WPT systems; and developed a flying-capacitor voltage balancing method without relying on current measurement, capable of withstanding dynamic step responses.
- Build a simulation model of the proposed WPT system using MATLAB Simulink to validate the A-SVPFM modulation strategy and the closed-loop voltage balancing control of the flying capacitor.
- Utilized SiC MOSFETs for converter in hardware experiment system, controlled by DSP28377D, designed PCB integrated controller and converter using Altium Designer. Successfully validated at a 150V operating voltage with frequency over 100kHz, achieving the anticipated performance metrics and system stability.
- Tested experiment system using oscilloscope, DC power supply, electronic load, and power analyzer.
- Key achievements include deriving a closed-form ZVS condition and implementing a sensor-less voltage balancing algorithm to realize wide-range soft-switching without circuit modifications, also designed and made an GaN prototype for future improvement.

Relevant Research Skills

- Research experience of DC-DC converter modeling; Modulation strategy design and modeling; Simulation Experiments via MATLAB Simulink
- PCB design via Altium Designer; Experience of IGBT application (SiC & GaN)
- Experience of DSP programming and debugging(TI DSP28377) in CCS and ESP32 programming in C
- Testing equipment: oscilloscope, DC power supply, electronic load, power analyzer, Handheld thermal imager